Iterated Function Systems in Approximating Fractal Dimensions

An ISI Math Club talk

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27/10/2025

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Motivation

Introduction

- Fractals appear everywhere in nature - in coastlines, cabbages, clouds and crystals
- They have details at infinitesimal scales - details won't vanish as you zoom in on a fractal
- They are often characterized by fractional dimensions, as we will see later
- I will not be providing a precise definition of fractals as it requires an extensive study of measure and dimension theory



A Computer-Generated 3D Fractal

Fractals in Nature



Fractals in Nature








Some Basic Definitions

Contractions and Contracting Similarities

Setup. Let $D \subset \mathbb{R}^n$ be closed. A map $S : D \to D$ is a *contraction* if there exists $0 \leq c < 1$ such that

$$|S(x) - S(y)| \leq c|x - y| \ \forall \ x, y \in D.$$

If moreover S satistifes $|S(x) - S(y)| = c|x - y| \ \forall \ x, y \in D$, we call S a *contracting similarity* with ratio c.

Iterated Function Systems

Definition 1. *An Iterated Function System (IFS) on D is a finite collection $\{S_1, \ldots, S_m\}$ of contractions $S_i : D \to D$.*

We often denote the IFS by $\mathcal{S} = \{S_i\}_{i=1}^{m}$.

Attractor of an IFS

Definition 2. *A nonempty compact set $F \subset D$ is an attractor (or invariant set) of the IFS $\mathcal{S}$ if*

$$F = \bigcup_{i=1}^{m} S_i(F).$$

Intuitively, F is made of scaled copies of itself, which leads us to guess that F will have fractal nature.

Hausdorff Metric

Setup. Let (X, d) be a metric space and denote by $\mathcal{K}(X)$ the collection of nonempty compact subsets of X. For $A, B \in \mathcal{K}(X)$ define the distance from a point to a set by

$$\operatorname{dist}(x, B) := \inf_{y \in B} d(x, y).$$

Definition 3. *The Hausdorff distance between A and B is*

$$d_H(A, B) := \max\left\{ \sup_{a \in A} \operatorname{dist}(a, B),\ \sup_{b \in B} \operatorname{dist}(b, A) \right\}.$$

Equivalently,

$$d_H(A, B) = \inf\{r > 0 :\ A \subset B^{(r)} \text{ and } B \subset A^{(r)}\},$$

where $B^{(r)} := \bigcup_{y \in B} B(y, r)$ is the r-neighborhood of B.

Iterated Function Theorem

Iterated Function Theorem (Existence and Uniqueness)

Theorem. Let $\{S_1, \ldots, S_m\}$ be contractions on a complete metric space (D closed in $\mathbb{R}^n$). Then there exists a unique nonempty compact set F such that

$$F = \bigcup_{i=1}^{m} S_i(F).$$

Moreover, if E is any nonempty compact set with $S_i(E) \subset E$ for all i, then

$$F = \bigcap_{k=1}^{\infty} S^k(E),$$

where $S(E) := \bigcup_{i=1}^{m} S_i(E)$ and S^k denotes k-fold composition.

Sketch of proof: Apply the Banach fixed point theorem to the transformation $\mathcal{T}(A) = \bigcup_i S_i(A)$ on the space of nonempty compact subsets with the Hausdorff metric.

Computing the Attractor

Two practical methods:

1. **Deterministic iteration**: Start with a compact set E with $S_i(E) \subset E$, iterate $E_{k+1} = \bigcup_i S_i(E_k)$. The attractor is the limit of this iteration.
2. **Chaos game (random iteration)**: Start with $x_0 \in D$, choose indices i_k at random (often uniformly) and set $x_{k+1} = S_{i_k}(x_k)$. The orbit $\bigcup_{k=0}^{\infty} x_k$ approximates the attractor. This is a much more practical method and is mainly used in the simulation of fractals.

Cantor Set

Cantor Set as an Attractor

Let $D = [0, 1] \subset \mathbb{R}$. Consider the two contracting similarities

$$S_1(x) = \frac{x}{3}, \qquad S_2(x) = \frac{x}{3} + \frac{2}{3},$$

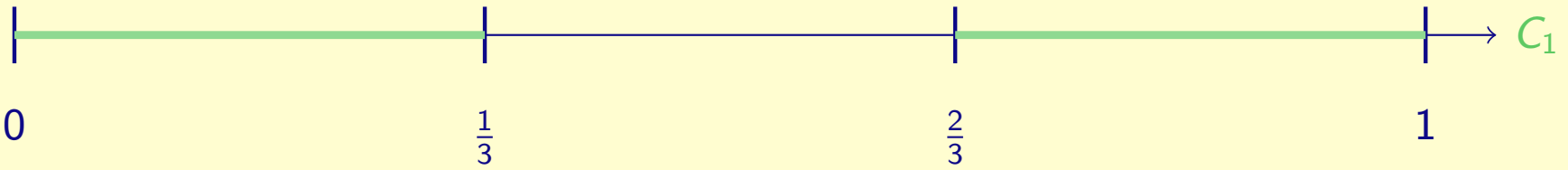
each with similarity ratio $c_1 = c_2 = \frac{1}{3}$. The collection $\mathcal{S} = \{S_1, S_2\}$ is an Iterated Function System (IFS) on D. The Cantor Set is the unique attractor of this IFS.

Cantor Set: Deterministic Construction

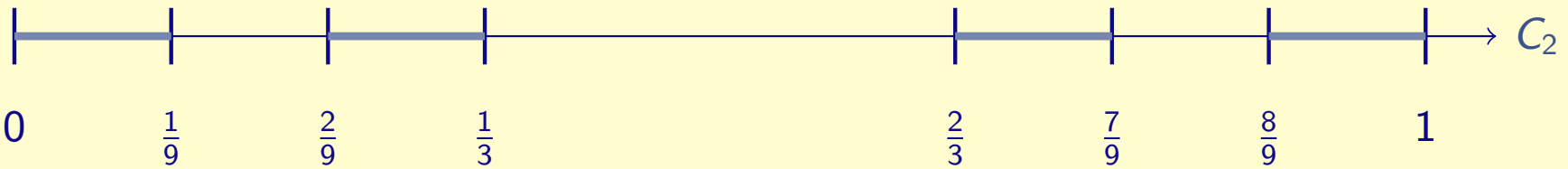
Start with the interval $E = [0, 1]$ and call it C_0.



Apply S_1 and S_2 on E to get $C_1 := S_1(E)\left(= \left[0, \frac{1}{3}\right]\right) \cup S_2(E)\left(= \left[\frac{2}{3}, 1\right]\right)$.



Apply S_1 and S_2 again to get $C_2 := S_1^2(E) \cup S_1 \circ S_2(E) \cup S_2 \circ S_1(E) \cup S_2^2(E)$



Sierpiński Carpet

Sierpiński Carpet as an IFS

Iterated Function System. Define 8 contracting similarities $S_i : \mathbb{R}^2 \to \mathbb{R}^2$ by

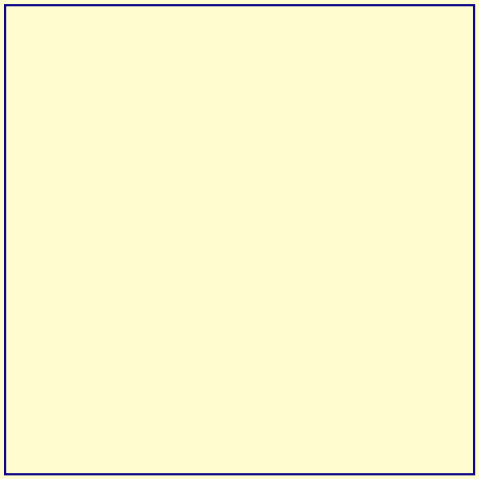
$$S_i(x) = \frac{1}{3}x + t_i, \qquad i = 1, \ldots, 8,$$

where each translation vector

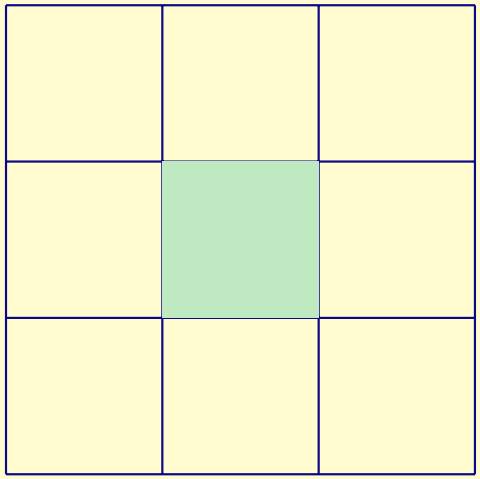
$$t_i \in \left\{(x, y) | x, y \in \left\{0, \frac{1}{3}, \frac{2}{3}\right\}\right\} \setminus \left(\frac{1}{3}, \frac{1}{3}\right).$$

corresponds to one of the retained squares. The Sierpiński carpet is the unique attractor of this IFS.

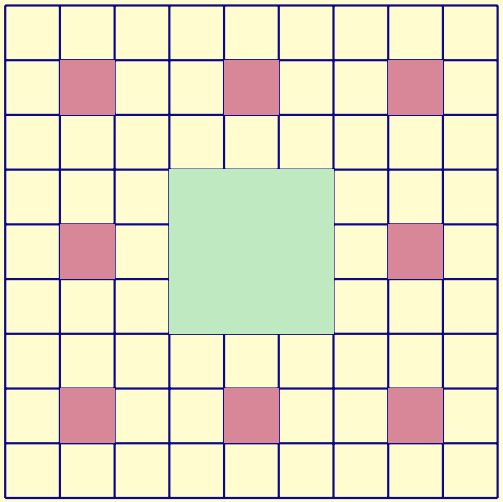
Construction of the Sierpiński Carpet



S_0



S_1



S_2

Sierpiński Carpet: Construction using Chaos Game

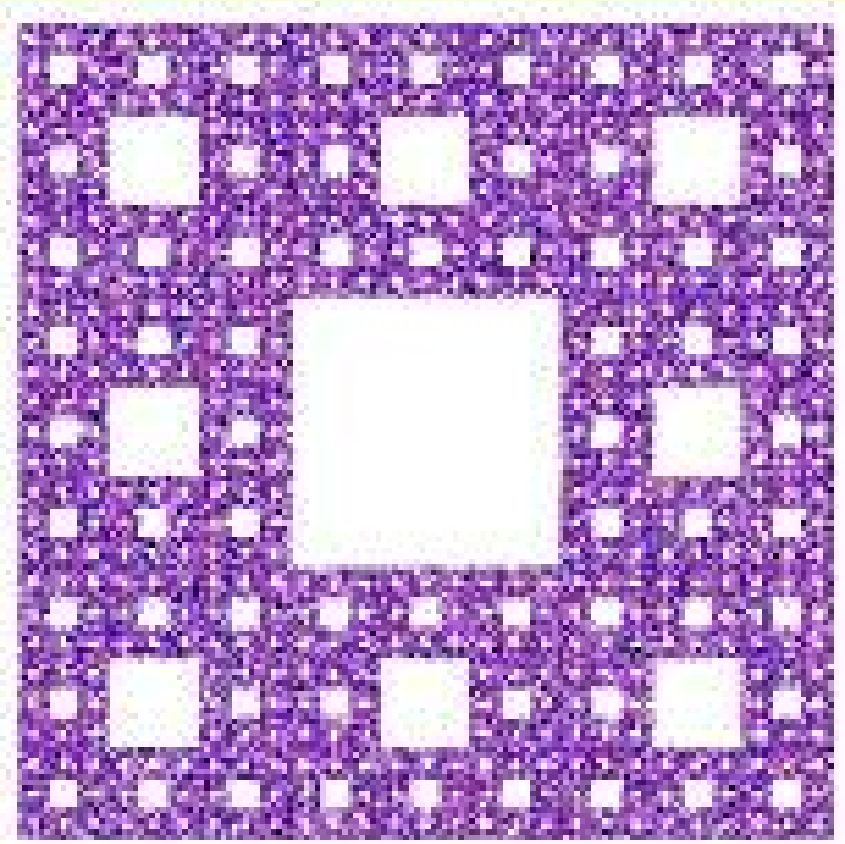


Figure 1: Sierpiński Carpet generated in R

Sierpiński Gasket

Sierpiński Gasket as an IFS

Iterated Function System. Define 3 contracting similarities $S_i : \mathbb{R}^2 \to \mathbb{R}^2$ by

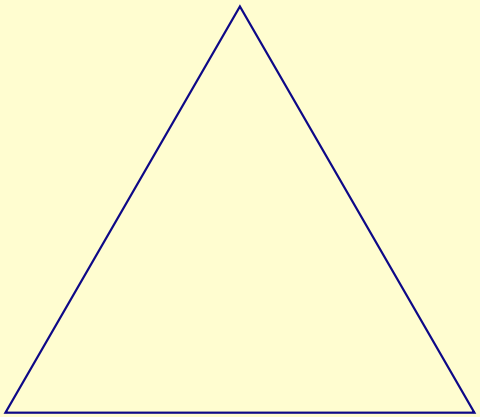
$$S_i(x) = \frac{1}{2}x + t_i, \qquad i = 1, 2, 3,$$

where each translation vector t_i corresponds to one of the vertices of an equilateral triangle, i.e.

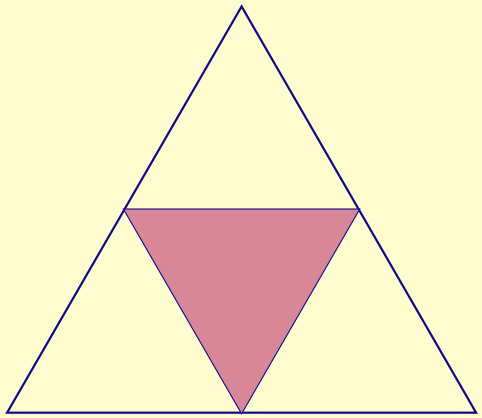
$$t_1 = (0, 0), \quad t_2 = \left(\frac{1}{2}, 0\right), \quad t_3 = \left(\frac{1}{4}, \frac{\sqrt{3}}{4}\right).$$

The Sierpiński gasket is the unique attractor of this IFS.

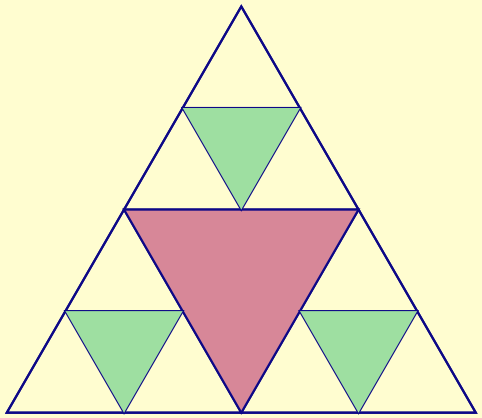
Construction of the Sierpiński Gasket



T_0



T_1



T_2

Sierpiński Gasket: Construction using Chaos Game

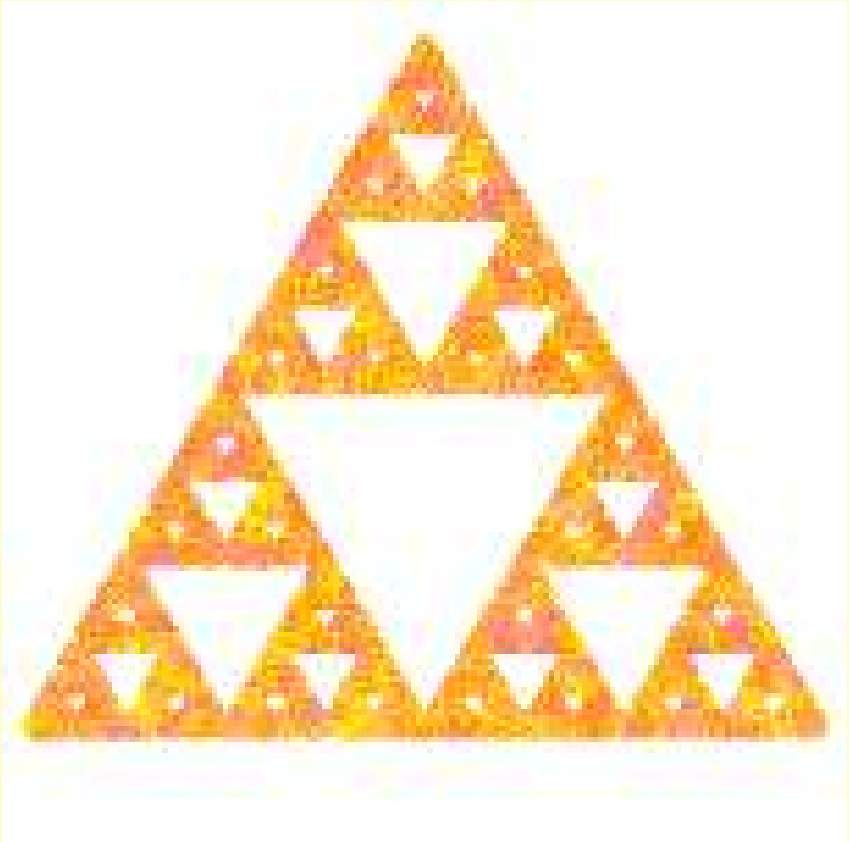


Figure 2: Sierpiński Gasket generated in R

Box Dimension

Box (Minkowski) Dimension

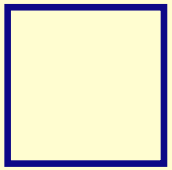
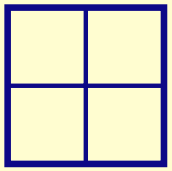
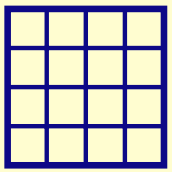
For a bounded set $F \subset \mathbb{R}^n$, let $N(\varepsilon)$ be the smallest number of cubes of side length ε required to cover F. The *box dimension* is

$$\dim_B(F) := \lim_{\varepsilon \to 0} \frac{\log N(\varepsilon)}{-\log \varepsilon},$$

when the limit exists. If the limit does not exist, we use lim sup and lim inf to defined the upper and lower box dimensions respectively.

Box Counting on $[0,1]^2$

Idea. To estimate the box-counting dimension of the unit square $[0,1]^2$, we cover it with squares of side length ϵ and count the minimum number of boxes required, denoted $N(\epsilon)$.

ϵ	$N(\epsilon)$	$\frac{\log N(\epsilon)}{\log(1/\epsilon)}$	Covering
1	1	–	
$\frac{1}{2}$	4	2	
$\frac{1}{4}$	16	2	
$\frac{1}{2^n}$	2^{2n}	2	–

Computation.

$$\frac{\log N(1/2^n)}{\log 2^n} = \frac{\log 2^{2n}}{\log 2^n} = \frac{2n \log 2}{n \log 2} = 2.$$

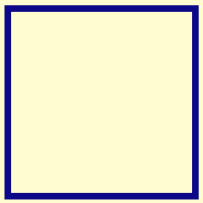
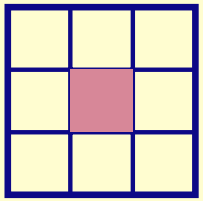
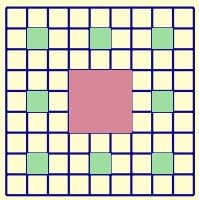
General case.

$$N(\epsilon) = \left\lceil \frac{1}{\epsilon^2} \right\rceil \quad \Rightarrow \quad \lim_{\epsilon \to 0} \frac{\log N(\epsilon)}{\log(1/\epsilon)} = 2.$$

Hence, $\dim_B([0,1]^2) = 2$.

Box Counting on the Sierpiński Carpet

Idea. Let $\epsilon > 0$ and let $N_n(\epsilon)$ be the minimal number of ϵ-squares needed to cover S_n (the n-th approximation of the carpet).

ϵ	$N_n(\epsilon)$	$\dfrac{\log N_n(\epsilon)}{\log(1/\epsilon)}$	Covering
1	1	–	
$\dfrac{1}{3}$	8	$\dfrac{\log 8}{\log 3}$	
$\dfrac{1}{9}$	64	$\dfrac{\log 8}{\log 3}$	
$\dfrac{1}{3^n}$	8^n	$\dfrac{\log 8}{\log 3}$	–

Computation.

$$\frac{\log N_n(3^{-n})}{\log 3^n} = \frac{\log 8^n}{\log 3^n} = \frac{n \log 8}{n \log 3} = \log_3 8.$$

General case. For general small ϵ and large n one can show

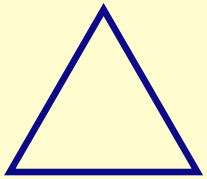
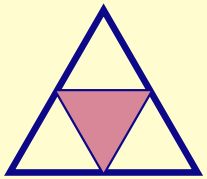
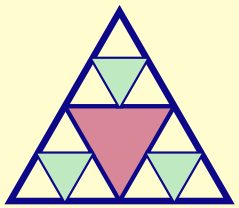
$$N_n(\epsilon) \asymp \frac{(8/9)^n}{\epsilon^2} \quad \text{and hence}$$

$$\lim_{\epsilon \to 0} \frac{\log\left(\lim_{n\to\infty} N_n(\epsilon)\right)}{\log(1/\epsilon)} = \log_3 8.$$

Consequently, $\dim_B(S) = \log_3 8$.

Box Counting on the Sierpiński Gasket

Idea. Let $\epsilon > 0$ and let $N_n(\epsilon)$ be the minimal number of ϵ-triangles needed to cover T_n (the n-th approximation of the gasket).

ϵ	$N_n(\epsilon)$	$\dfrac{\log N_n(\epsilon)}{\log(1/\epsilon)}$	Covering
1	1	–	
$\dfrac{1}{2}$	3	$\dfrac{\log 3}{\log 2}$	
$\dfrac{1}{4}$	9	$\dfrac{\log 3}{\log 2}$	
$\dfrac{1}{2^n}$	3^n	$\dfrac{\log 3}{\log 2}$	–

Computation.

$$\frac{\log N_n(2^{-n})}{\log 2^n} = \frac{\log 3^n}{\log 2^n} = \frac{n \log 3}{n \log 2} = \log_2 3.$$

General case. For general small ϵ and large n one can show

$$N_n(\epsilon) \asymp \frac{(3/4)^n}{\epsilon^2} \quad \text{and hence}$$

$$\lim_{\epsilon \to 0} \frac{\log\left(\lim_{n\to\infty} N_n(\epsilon)\right)}{\log(1/\epsilon)} = \log_2 3.$$

Consequently, $\dim_B(T) = \log_2 3$.

Hausdorff Dimension

Hausdorff Dimension

For $s \geq 0$, define the s-dimensional Hausdorff measure

$$\mathcal{H}^s(F) = \lim_{\delta \downarrow 0} \inf \left\{ \sum_i (\operatorname{diam} U_i)^s : \ F \subset \bigcup_i U_i, \ \operatorname{diam} U_i < \delta \right\}.$$

Then the Hausdorff dimension is

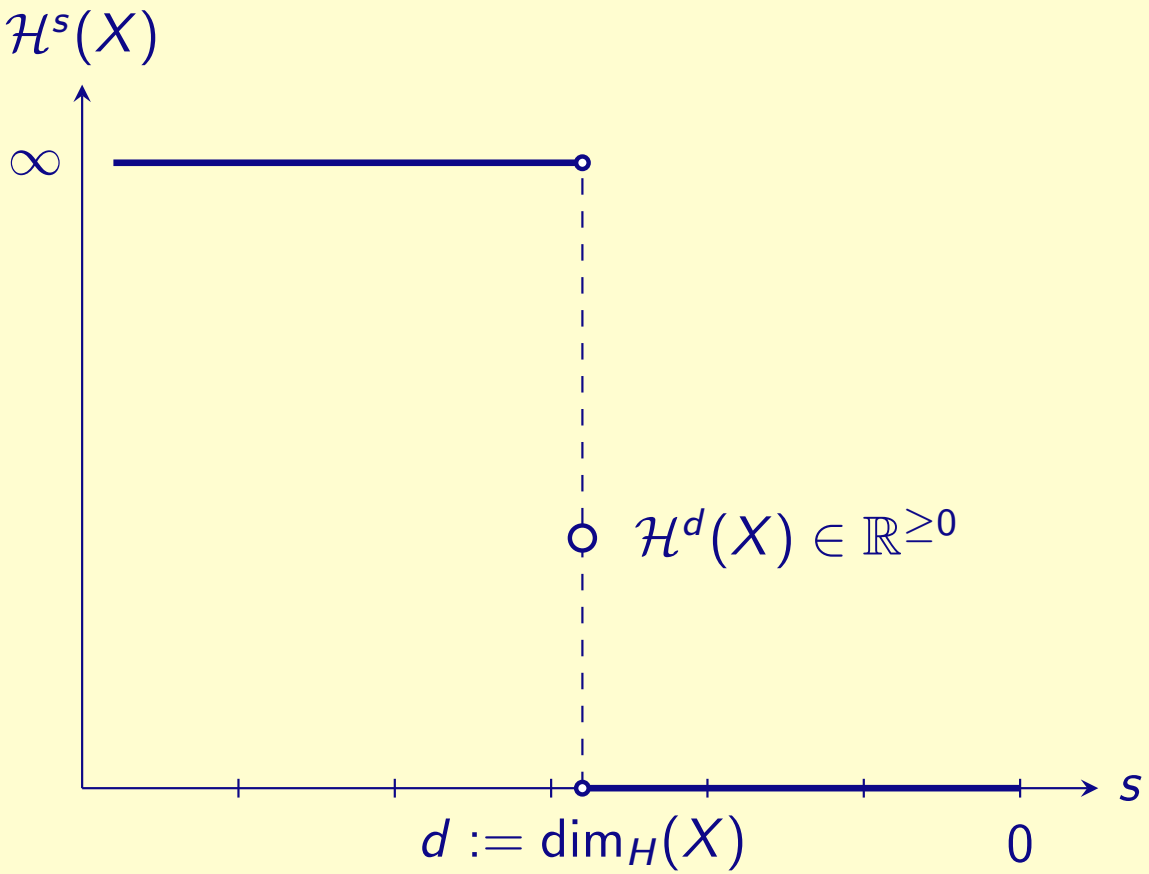
$$\dim_H(F :) = \inf\{s \geq 0 : \ \mathcal{H}^s(F) = 0\}.$$

If $\dim_H(F) \neq 0$,

$$\dim_H(F) = \sup\{s \geq 0 : \ \mathcal{H}^s(F) = \infty\}.$$

Graphical Interpretation of Hausdorff Dimension

Visualize the function $s \mapsto \sum_i (\text{diam } U_i)^s$ for an optimal cover. As s increases, this sum decreases; the critical exponent where it transitions from ∞ to 0 is the Hausdorff dimension.



Hausdorff Dimension of an Attractor

$F = \bigcup_{i=1}^{m} S_i(F)$ be an attractor with the union disjoint and say each S_i is a similarity with ratio c_i, then

$$\mathcal{H}^s(F) = \sum_{i=1}^{m} \mathcal{H}^s(S_i(F)) = \sum_{i=1}^{m} c_i^s \, \mathcal{H}^s(F).$$

If $0 < \mathcal{H}^s(F) < \infty$, s is the Hausdorff dimension of F and cancellation yields

$$\sum_{i=1}^{m} c_i^{\dim_H(F)} = 1.$$

Open Set Condition (OSC)

However this only works if the union $\bigcup_{i=1}^{m} S_i(F)$ is disjoint. We will now introduce a condition called the **open set condition** which ensures that the above property still holds.

The IFS $\{S_i\}_{i=1}^{m} : D \mapsto D$ satisfies the OSC if there exists a nonempty bounded open set $U \subseteq D$ such that

$$\bigcup_{i=1}^{m} S_i(U) \subset U \quad \text{and} \quad S_i(U) \cap S_j(U) = \emptyset \text{ for } i \neq j.$$

Hausdorff Dimension of an Attractor under OSC

Theorem. Suppose the IFS $\{S_i\}_{i=1}^m$ consists of contracting similarities with ratios c_i and satisfies the OSC. Let s be the unique solution of

$$\sum_{i=1}^{m} c_i^s = 1.$$

Then the attractor F of the IFS satisfies

$$\dim_H(F) = \dim_B(F) = s,$$

and $0 < \mathcal{H}^s(F) < \infty$.

Generalization for Contractions

Proposition. 1. *Let F be the attractor of an IFS consisting of contractions $\{S_1, \ldots, S_m\}$ on a closed subset $D \subseteq \mathbb{R}^n$ such that*

$$|S_i(x) - S_i(Y)| \leq c_i|x - y| \quad (x, y \in D)$$

with $0 < c_i < 1$ for each i. Then

$$\dim_H(F) \leq s \text{ and } \overline{\dim_B}(F) \leq s, \text{ where } \sum_{i=1}^{m} c_i^s = 1.$$

Generalization for Contractions

Proposition. 2. *Consider the IFS consisting of contractions* $\{S_1, \ldots, S_m\}$ *on a closed subset* D *of* $\mathbb{R}^n$ *such that*

$$b_i|x - y| \leq |S_i(x) - S_i(y)| \qquad (x, y \in D)$$

with $0 < b_i < 1$ *for each* i*. Assume that the (non-empty compact) attractor* F *satisfies*

$$F = \bigcup_{i=1}^{m} S_i(F),$$

with this union disjoint. Then F *is totally disconnected and*

$$\dim_H F \geq s \quad \textit{where} \quad \sum_{i=1}^{m} b_i^s = 1.$$

Examples

Examples

1. **Cantor Set:** The similarity ratios for the IFS correcponding to the Cantor Set are $c_1 = c_2 = \frac{1}{3}$. OSC is satisfied by the interval $[0, 1]$:

$$\dim_H(C) = \dim_B(C) = d \text{ s.t. } \frac{1}{3}^d + \frac{1}{3}^d = 1 \implies d = \log_3(2) \approx 0.631.$$

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2. **Sierpiński Carpet:** The similarity ratios for the IFS correcponding to the Sierpiński Carpet are $c_i = \frac{1}{3}\ \forall\ 1 \leq i \leq 8$. OSC is satisfied by the box $[0, 1]^2$:

$$\dim_H(S) = \dim_B(S) = d \text{ s.t. } 8 \cdot \frac{1}{3}^d = 1 \implies d = \log_3(8) \approx 1.893.$$

Examples

1. **Cantor Set:** The similarity ratios for the IFS correcponding to the Cantor Set are $c_1 = c_2 = \frac{1}{3}$. OSC is satisfied by the interval $[0, 1]$:

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2. **Sierpiński Carpet:** The similarity ratios for the IFS correcponding to the Sierpiński Carpet are $c_i = \frac{1}{3}\ \forall\ 1 \leq i \leq 8$. OSC is satisfied by the box $[0, 1]^2$:

$$\dim_H(S) = \dim_B(S) = d \text{ s.t. } 8 \cdot \frac{1}{3}^d = 1 \implies d = \log_3(8) \approx 1.893.$$

3. **Sierpiński Gasket:** The similarity ratios for the IFS correcponding to the Sierpiński Gasket are $c_1 = c_2 = c_3 = \frac{1}{2}$. OSC is satisfied by the triangle $(0, 0) - (1, 0) - \left(\frac{1}{2}, \frac{\sqrt{3}}{2}\right)$:

$$\dim_H(T) = \dim_B(T) = d \text{ s.t. } 3 \cdot \frac{1}{2}^d = 1 \implies d = \log_2(3) \approx 1.585.$$

"Non-linear" Cantor Set

Let $D = \left[\frac{1}{2}(1+\sqrt{3}), 1+\sqrt{3}\right]$ and let $S_1, S_2 : D \mapsto D$ be given by

$$S_1(x) = 1 + \frac{1}{x} \quad S_2(x) = 2 + \frac{1}{x}.$$

Then

$$0.44 < \dim_H(F) \leq \underline{\dim_B} F \leq \overline{\dim_B}(F) < 0.66$$

where F is the attractor of $\{S_1, S_2\}$. In fact,

$$\dim_H(F) = 0.531.$$

"Non-Linear" Cantor Set

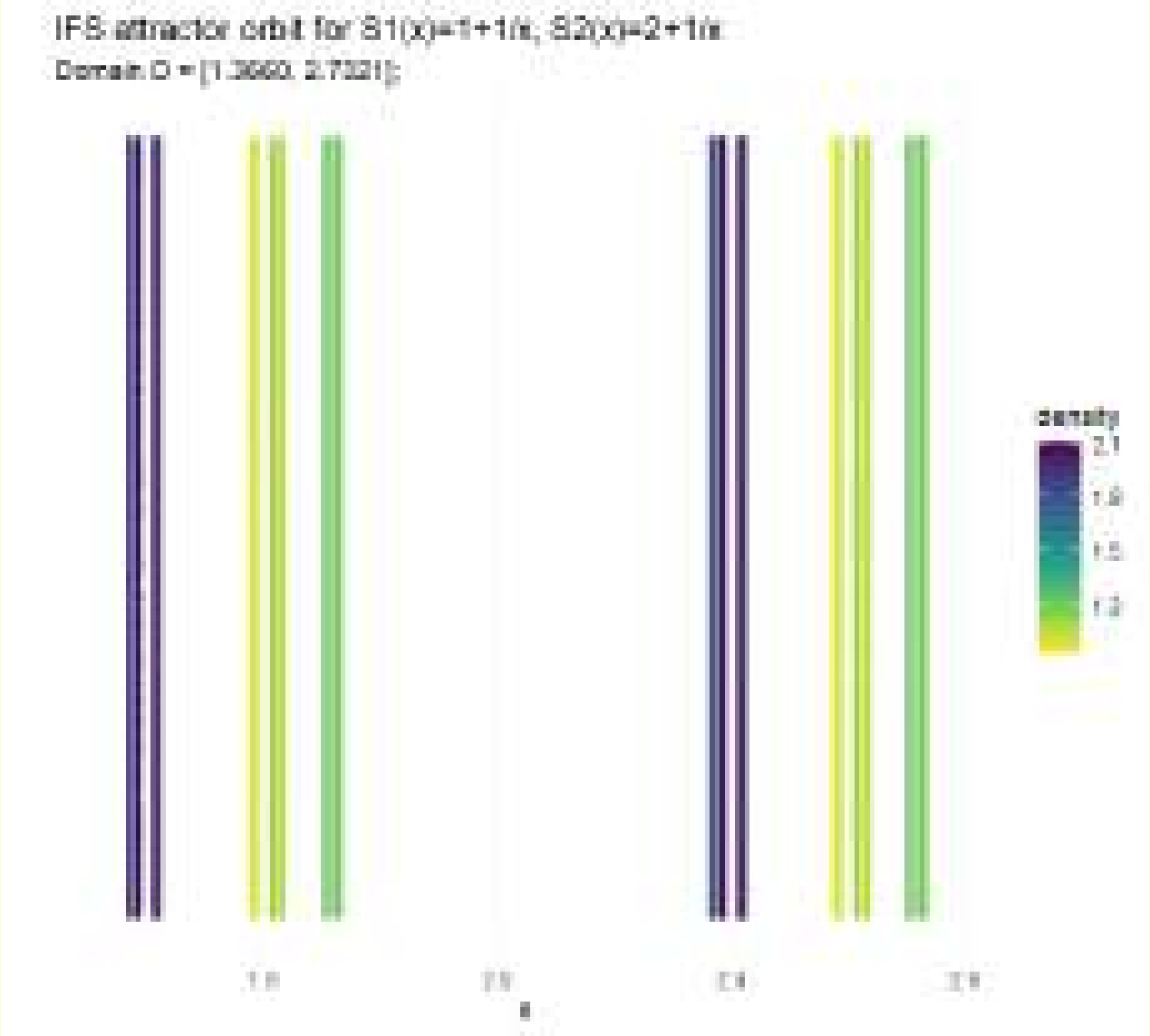


Figure 3: "Non-Linear Cantor Set generated on R"

Self-Affine Sets

Self-affine Transformations

A map $T : \mathbb{R}^n \to \mathbb{R}^n$ is *affine* if $T(x) = Ax + t$ where A is a non-singular linear map and $t \in \mathbb{R}^n$. If A has eigenvalues with different magnitudes, T is *non-uniformly contracting* and sets generated by such maps are called self-affine.

An affine transformation is a combination of a translation, rotation, dilation and a reflection.

Self-affine Sets

A compact set F satisfying

$$F = \bigcup_{i=1}^{m} (A_i F + t_i)$$

for affine contractions $A_i x + t_i$ is called *self-affine*.

"Discontinuous" Dimension?

Theorem 4. *Let S_1, S_2 be the affine contractions on $\mathbb{R}^2$ that map the unit square onto the rectangles R_1 and R_2 of sides $1 \times \epsilon$ where $0 < \epsilon < \frac{1}{2}$. The rectangle R_1 abuts the y-axis, but the end of R_2 is distance $0 \leq \lambda \leq \frac{1}{2}$ from the y-axis. If F is the attractor of $\{S_1, S_2\}$, we have*

$$\dim_H(F) \begin{cases} \geq 1, & \lambda > 0 \\ = -\log_\epsilon 2 < 1, & \lambda = 0 \end{cases}.$$

Discontinuity of the Dimension of Self-Affine Sets

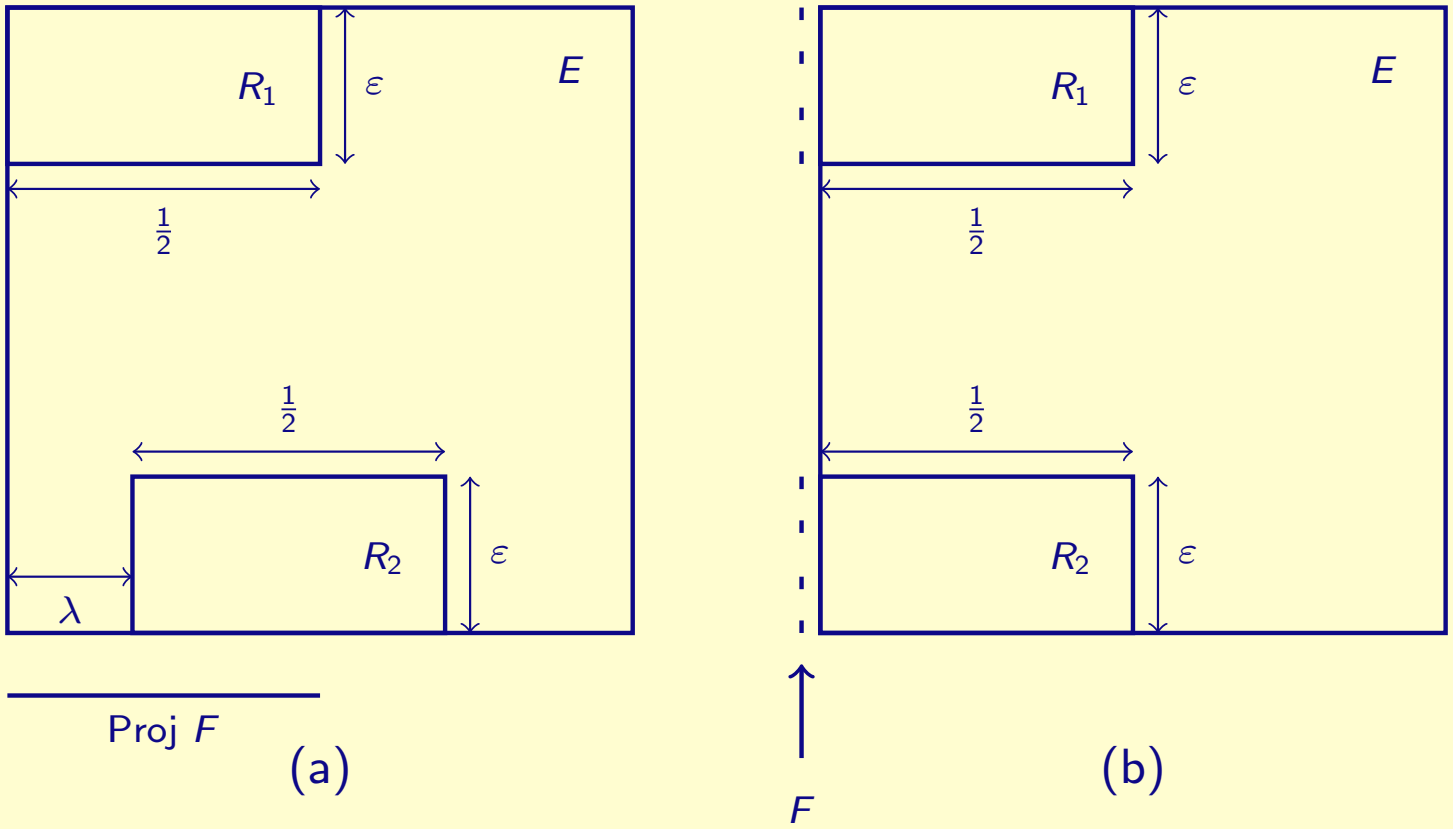


Figure 4: Discontinuity of the dimension of self-affine sets

Is $\dim_B = \dim_H$?

Theorem 5. *Let the unit square E_0 be divided into a $p \times q$ array of rectangles of sides $\frac{1}{p}$ and $\frac{1}{q}$, where $p < q \in \mathbb{N}$. Select a subcollection of these rectangles to form E_1, and let N_j denote the number of rectangles selected from the jth column for $1 \leq j \leq p$. Iterate this construction in the usual way and let F be the limiting set obtained. Then*

$$\dim_{\mathrm{H}} F = \frac{1}{\log p} \log \left(\sum_{j=1}^{p} N_j^{\log p / \log q} \right)$$

and

$$\dim_{\mathrm{B}} F = \frac{\log p_1}{\log p} + \frac{1}{\log q} \log \left(\frac{1}{p_1} \sum_{j=1}^{p} N_j \right),$$

p_1 being the number of columns containing at least one rectangle of E_1.

Is $\dim_B = \dim_H$?

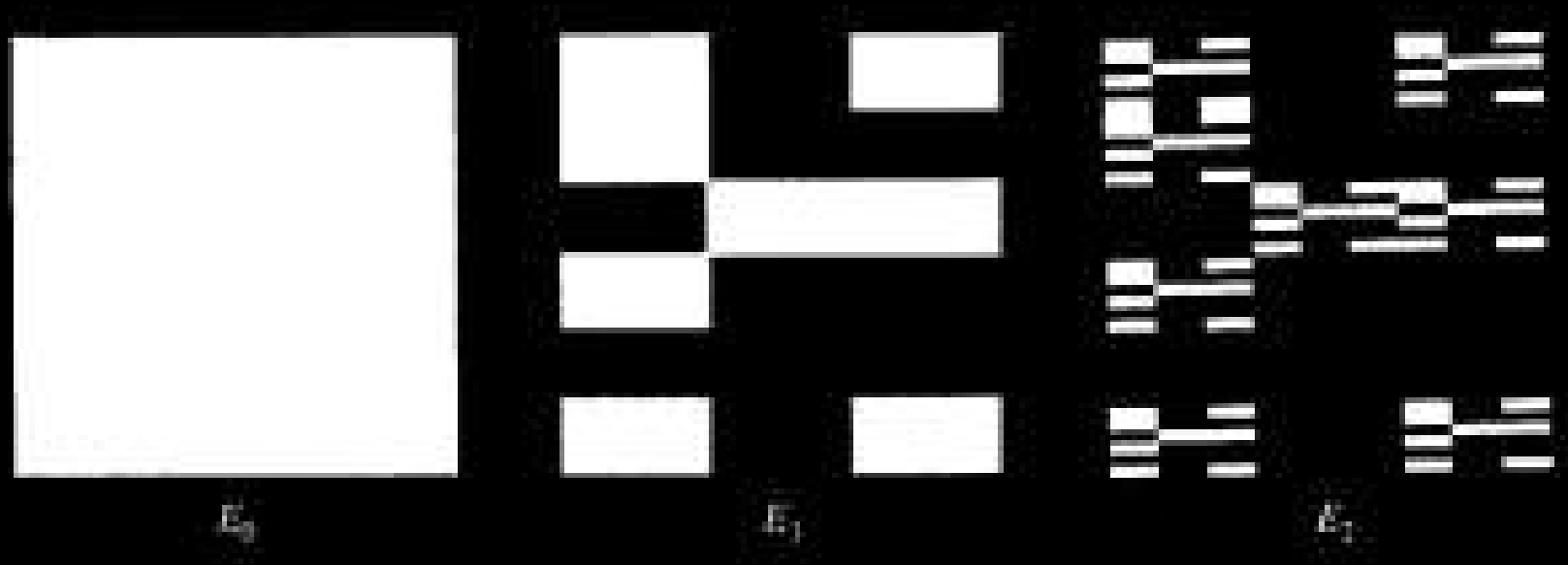


Figure 5: A $p = 3$ and $q = 6$ case of the above example

References


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A. Chatterjee and S. Athreya, *Fractals: An Understanding via Topological, Box, Similarity and Hausdorff Dimensions*, 2025.

Thank You

THANK YOU

You may check my S.N. Bhatt report on Fractal Dimensions in order to study the theory more rigorously. Please mail me at bmat2303@isibang.ac.in for the slides and the report, and also if you want to discuss further.